nudging our behavior, they will become more capable at manipulating us. We will now discuss how AIs could be leveraged by malicious actors to create a fractured and dysfunctional society.

AIs could pollute the information ecosystem with motivated lies. Sometimes ideas spread not because they are true, but because they serve the interests of a particular group. "Yellow journalism" was coined as a pejorative reference to newspapers that advocated war between Spain and the United States in the late 19th century, because they believed that sensational war stories would boost their sales [20]. When public information sources are flooded with falsehoods, people will sometimes fall prey to lies, or else come to distrust mainstream narratives, both of which undermine societal integrity.

Unfortunately, AIs could escalate these existing problems dramatically. First, AIs could be used to generate unique, personalized disinformation at a large scale. While there are already many social media bots [21], some of which exist to spread disinformation, historically they have been run by humans or primitive text generators. The latest AI systems do not need humans to generate personalized messages, never get tired, and could potentially interact with millions of users at once [22].

AIs can exploit users' trust. Already, hundreds of thousands of people pay for chatbots marketed as lovers and friends [23], and one man's suicide has been partially attributed to interactions with a chatbot [24]. As AIs appear increasingly human-like, people will increasingly form relationships with them and grow to trust them. AIs that gather personal information through relationship-building or by accessing extensive personal data, such as a user's email account or personal files, could leverage that information to enhance persuasion. Powerful actors that control those systems could exploit user trust by delivering personalized disinformation directly through people's "friends."

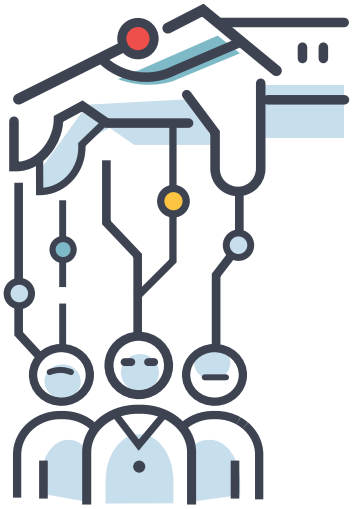

Figure 4: AIs will enable sophisticated personalized influence campaigns that may destabilize our shared sense of reality.

AIs could centralize control of trusted information. Separate from democratizing disinformation, AIs could centralize the creation and dissemination of trusted information. Only a few actors have the technical skills and resources to develop cutting-edge AI systems, and they could use these AIs to spread their preferred narratives. Alternatively, if AIs are broadly accessible this could lead to widespread disinformation, with people retreating to trusting only a small handful of authoritative sources [25]. In both scenarios, there would be fewer sources of trusted information and a small portion of society would control popular narratives.

AI censorship could further centralize control of information. This could begin with good intentions, such as using AIs to enhance fact-checking and help people avoid falling prey to false narratives. This would not necessarily solve the problem, as disinformation persists today despite the presence of fact-checkers.

Even worse, purported "fact-checking AIs" might be designed by authoritarian governments and others to suppress the spread of true information. Such AIs could be designed to correct most common misconceptions but provide incorrect information about some sensitive topics, such as human rights violations committed by certain countries. But even if fact-checking AIs work as intended, the public might eventually become entirely dependent on them to adjudicate the truth, reducing people's autonomy and making them vulnerable to failures or hacks of those systems.

In a world with widespread persuasive AI systems, people's beliefs might be almost entirely determined by which AI systems they interact with most. Never knowing whom to trust, people could retreat even further into ideological enclaves, fearing that any information from outside those enclaves might be a sophisticated lie. This would erode consensus reality, people's ability to cooperate with others, participate in civil society, and address collective action problems. This would also reduce our ability to have a conversation as a species about how to mitigate existential risks from AIs.

In summary, AIs could create highly effective, personalized disinformation on an unprecedented scale, and could be particularly persuasive to people they have built personal relationships with. In the hands of many people, this could create a deluge of disinformation that debilitates human society, but, kept in the hands of a few, it could allow governments to control narratives for their own ends.

2.4 Concentration of Power

We have discussed several ways in which individuals and groups might use AIs to cause widespread harm, through bioterrorism; releasing powerful, uncontrolled AIs; and disinformation. To mitigate these risks, governments might pursue intense surveillance and seek to keep AIs in the hands of a trusted minority. This reaction, however, could easily become an overcorrection, paving the way for an entrenched totalitarian regime that would be locked in by the power and capacity of AIs. This scenario represents a form of "top-down" misuse, as opposed to "bottom-up" misuse by citizens, and could in extreme cases culminate in an entrenched dystopian civilization.

Figure 5: Ubiquitous monitoring tools, tracking and analyzing every individual in detail, could facilitate the complete erosion of freedom and privacy.

AIs could lead to extreme, and perhaps irreversible concentration of power. The persuasive abilities of AIs combined with their potential for surveillance and the advancement of autonomous weapons could allow small groups of actors to "lock-in" their control over society, perhaps permanently. To operate effectively, AIs require a broad set of infrastructure components, which are not equally distributed, such as data centers, computing power, and big data. Those in control of powerful systems may use them to suppress dissent, spread propaganda and disinformation, and otherwise advance their goals, which may be contrary to public wellbeing.

Figure 6: If material control of AIs is limited to few, it could represent the most severe economic and power inequality in human history.

AIs may entrench a totalitarian regime. In the hands of the state, AIs may result in the erosion of civil liberties and democratic values in general. AIs could allow totalitarian governments to efficiently collect, process, and act on an unprecedented volume of information, permitting an ever smaller group of people to surveil and exert complete control over the population without the need to enlist millions of citizens to serve as willing government functionaries. Overall, as power and control shift away from the public and toward elites and leaders, democratic governments are highly vulnerable to totalitarian backsliding. Additionally, AIs could make totalitarian regimes much longer-lasting; a major way in which such regimes have been toppled previously is at moments of vulnerability like the death of a dictator, but AIs, which would be hard to "kill," could provide much more continuity to leadership, providing few opportunities for reform.

AIs can entrench corporate power at the expense of the public good. Corporations have long lobbied to weaken laws and policies that restrict their actions and power, all in the service of profit. Corporations in control of powerful AI systems may use them to manipulate customers into spending more on their products even to the detriment of their own wellbeing. The concentration of power and influence that could be afforded by AIs could enable corporations to exert unprecedented control over the political system and entirely drown out the voices of citizens. This could occur even if creators of these systems know their systems are self-serving or harmful to others, as they would have incentives to reinforce their power and avoid distributing control.

In addition to power, locking in certain values may curtail humanity's moral progress. It's dangerous to allow any set of values to become permanently entrenched in society. For example, AI systems have learned racist and sexist views [26], and once those views are learned, it can be difficult to fully remove them. In addition to problems we know exist in our society, there may be some we still do not. Just as we abhor some moral views widely held in the past, people in the future may want to move past moral views that we hold today, even those we currently see no problem with. For example, moral defects in AI systems would be even worse if AI systems had been trained in the 1960s, and many people at the time would have seen no problem with that. We may even be unknowingly perpetuating moral catastrophes today [27]. Therefore, when advanced AIs emerge and transform the world, there is a risk of their objectives locking in or perpetuating defects in today's values. If AIs are not designed to continuously learn and update their understanding of societal values, they may perpetuate or reinforce existing defects in their decision-making processes long into the future.

In summary, although keeping powerful AIs in the hands of a few might reduce the risks of terrorism, it could further exacerbate power inequality if misused by governments and corporations. This could lead to totalitarian rule and intense manipulation of the public by corporations, and could lock in current values, preventing any further moral progress.

Story: Bioterrorism

The following is an illustrative hypothetical story to help readers envision some of these risks. This story is nonetheless somewhat vague to reduce the risk of inspiring malicious actions based on it.

A biotechnology startup is making waves in the industry with its AI-powered bioengineering model. The company has made bold claims that this new technology will revolutionize medicine through its ability to create cures for both known and unknown diseases. The company did, however, stir up some controversy when it decided to release the program to approved researchers in the scientific community. Only weeks after its decision to make the model open-source on a limited basis, the full model was leaked on the internet for all to see. Its critics pointed out that the model could be repurposed to design lethal pathogens and claimed that the leak provided bad actors with a powerful tool to cause widespread destruction, opening it up to abuse without safeguards in place.

Unknown to the public, an extremist group has been working for years to engineer a new virus designed to kill large numbers of people. Yet given their lack of expertise, these efforts have so far been unsuccessful. When the new AI system is leaked, the group immediately recognizes it as a potential tool to design the virus and circumvent legal and monitoring obstacles to obtain the necessary raw materials. The AI system successfully designs exactly the kind of virus the extremist group was hoping for. It also provides step-by-step instructions on how to synthesize large quantities of the virus and circumvent any obstacles to spreading it. With the synthesized virus in hand, the extremist group devises a plan to release the virus in several carefully chosen locations in order to maximize its spread.

The virus has a long incubation period and spreads silently and quickly throughout the population for months. By the time it is detected, it has already infected millions and has an alarmingly high mortality rate. Given its lethality, most who are infected will ultimately die. The virus may or may not be contained eventually, but not before it kills millions of people.

2.5 Suggestions

We have discussed two forms of misuse: individuals or small groups using AIs to cause a disaster, and governments or corporations using AIs to entrench their influence. To avoid either of these risks being realized, we will need to strike a balance in terms of the distribution of access to AIs and governments' surveillance powers. We will now discuss some measures that could contribute to finding that balance.